

5	(a) (i) change in kinetic energy = $\frac{1}{2}mv^2$	C1	
	= $0.5 \times 25 \times (0.64)^2 = 5.1(2) \text{ J}$	A1	[2]
	(ii) zero	A1	[1]
	(iii) $(-)5.1(2) \text{ J}$	A1	[1]
	(b) (i) $PE = mgh$	C1	
	= $350 \times 0.64 \times 25$	C1	
	= 5600 J	A1	[3]
	<i>(If full length used allow 1/3)</i>		
	(ii) $P = Fv$ or gain in PE/t , E_p/t or work done $/t$, W/t	C1	
	= 350×0.64 or $5600 / 25$		
	= $220 (224) \text{ W}$	A1	[2]